

Rick Clinton

Independent AI Governance & Safety Researcher

Chennai, India | +91 9176933935 | clinpeter@gmail.com | github.com/Rick-Clinton-jpg

SUMMARY

Independent researcher building and publishing a deterministic trust-and-safety architecture for AI reasoning pipelines: input-stage manipulation detection, claim validation, structured deliberation, interaction-level trajectory tracking, and external agent-behavior auditing — evaluated against real model output and documented with honest limitations rather than only favorable findings. Began self-directed, AI-assisted development in July 2026; the six-repository research portfolio below was built in a concentrated nine-day span (Aug 6–15, 2026), followed by three applied builds delivered against hard external deadlines across separate hackathons and challenges in August 2026. Complements this with 5+ years directing enterprise program delivery at Walmart and Amazon, where research rigor, root-cause analysis, and stakeholder communication were exercised daily in quality- and SLA-driven environments.

CORE COMPETENCIES

Deterministic Safety Architecture • Claim & Evidence Validation • Evaluation Design (adversarial, schema-enforced) • Multi-Agent System Design • Prior-Art & Literature Review • Adversarial / Red-Team Testing • Technical Publishing

INDEPENDENT RESEARCH PORTFOLIO

Six-repository system built around one thesis: probabilistic models should not adjudicate their own trust boundaries — deterministic rules should, and AI agent behavior should be externally audited, not self-reported. Core pipeline: Sentry → Reasoning Kernel → Review Board, coordinated but never fused; extended by an interaction-level trajectory layer (IntentGraph) and an external agent-oversight tool (Warden).

Reasoning Kernel github.com/Rick-Clinton-jpg/reasoning-kernel — Aug 6, 2026

- Directed design and build of a deterministic 8-rule integrity layer for AI-generated reasoning graphs — facts without provenance are demoted to assumptions rather than deleted, and a conclusion cannot exceed the confidence of what supports it.
- Directed a 24-case evaluation built on invented entities so correct answers required reading the source rather than relying on prior knowledge; achieved 16/16 accuracy on resolvable cases and 8/8 honest “not enough information” abstentions against Claude Sonnet 4.6.
- Reviewed live evaluation results, identified two real defects in the kernel's own enforcement logic that 200 synthetic test cases had missed, and directed and verified their correction.

Sentry github.com/Rick-Clinton-jpg/Sentry — Aug 13, 2026

- Directed build of an offline, rule-based detector for prompt-injection and adversarial-input patterns — six detection rules, a 22-case pytest suite, and CI validated across three Python versions; packaged as an installable Claude Code plugin/skill.

Review Board github.com/Rick-Clinton-jpg/Review-Board — Aug 13, 2026

- Directed design of a six-stage structured deliberation protocol (Claude Code skill): restate objective, map every solution path, run an independently-worded second-pass prior-art check, score options against disclosed criteria, gather a six-angle review, and stop for explicit approval before implementation begins.
- Tested and deliberately abandoned an automatic-trigger design after adversarial testing showed it firing unreliably, rather than continuing to tune around the failure; redesigned as manual-invocation-only. Validated the review stage's ability to catch and correct a wrong score — not just confirm a correct one — via fault injection on two independent scenarios.

Trust Boundary github.com/Rick-Clinton-jpg/trust-boundary — Aug 13, 2026

- Directing the coordinator layer linking Sentry, Reasoning Kernel, and Review Board under a “coordinate, not fuse” principle — each tool stays inside the scope it was actually validated for; an earlier unified-graph design was deliberately rejected to preserve that boundary.
- While confirming integration points against each tool's actual source (not documented behavior), found and fixed two real defects: a type-check bug silently blocking Sentry's findings from ever triggering a rejection, and an assumed API that didn't match Reasoning Kernel's real interface.

IntentGraph (intent-layer) github.com/Rick-Clinton-jpg/intent-layer — Aug 14, 2026

- Directed build of a prototype trajectory-tracking layer that detects when a conversation returns to an objective that previously triggered a safety boundary — a failure mode structurally invisible to classifiers that evaluate only the current message in isolation.
- Evaluated on 15 synthetic trajectories: 100% detection / 0% false positives vs. 33% for a naive keyword baseline. Two hard constraints — confidence as a similarity multiplier, never a gate; no risk elevation without genuine prior boundary history — implemented and verified in code, with limitations documented honestly.

Warden github.com/Rick-Clinton-jpg/warden — Aug 15, 2026

- Directed build of an external observer for long-running AI coding agent sessions (Claude Code, Codex, and others) that does not control or interrupt the agent — it only compares the agent's self-reported status against its original stated objective, tags the result MATCH / DRIFT / UNCLEAR / DIVERGENT, and writes an append-only audit trail.
- Replaced an initial single-session polling design with a multi-session daemon after confirming the original approach registered a session but never actually re-checked it in the background.

Security Research

- Independently discovered and responsibly disclosed a Missing Authorization vulnerability in a third-party open-source AI agent framework (Microsoft MSRC-acknowledged; case resolved as Complete — Duplicate), identified using an AI-assisted source-review methodology consistent with the deterministic, externally-verified approach applied throughout this portfolio.

Paper — “The Gap and the Bridge” rick-clinton-jpg.github.io/papers/the-gap-and-the-bridge.html

- Published a working paper (Aug 2026) arguing AI systems hold no persistent model of their user, and that cross-vendor model divergence — a byproduct of market competition, not design — can function as a distributed error-correction resource; cited against current research on evolving user intent and multi-agent orchestration.

APPLIED BUILDS & COMPETITIONS (August 2026)

Three additional systems designed and shipped under external competition deadlines, applying the same verification-first architecture to new problem domains.

Quorum github.com/Rick-Clinton-jpg/Quorum — Google “All Things Agentic” Hackathon

- Directed design and deployment of a multi-agent enterprise fleet-coordination system to Google Cloud Run with Vertex AI and Firestore-backed persistence; published a live build-in-public development log.

Lex *EPFO claim-clarity tool* — Build What Moves India × OpenAI

- Directed design of a fully offline, single-file citizen-facing tool that explains EPFO Form 31/10C claim rejections in plain language across three user journeys, with bilingual (Hindi/English) support, redacted audit logging, and explainability shown on every verdict.

AI Kavach *Territorial Army / CyberPeace* — Terrier Cyber Quest 3.0 — Aug 2026

- Directed design of a lightweight, deterministic verification wrapper enabling any locally-run LLM to operate safely inside an air-gapped defense sandbox — applying the same “external proof over model self-report” architecture from the core research portfolio to a national-security context.

WORK EXPERIENCE

Project Management Specialist — Walmart, Chennai Aug 2022 – Jun 2025

- Directed a structured 8-week onboarding program for 15–22 international sellers/month, using discovery-call research to tailor each engagement; served as root-cause escalation point across engineering and catalog teams.
- Designed a duplicate-item identification process from scratch, standardizing key attributes into a playbook still in use today; awarded Stellar Stakeholder Recognition (April 2023).

Catalogue Associate — Amazon Pvt Ltd Apr 2019 – May 2021

- Managed catalog operations and vendor coordination, maintaining SLA and quality standards across a high-volume product set.

EDUCATION

B.Tech, Mechanical Engineering — SRM University

Diploma, Modern Applied Psychology (DiMAP)